Scaling Laws for Biological Realism in DNA Language Models

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1 Research Question and Motivation

How does the length over which generated DNA preserves natural genomic patterns change with model size? We will compare different-sized DNA language models to test whether larger models maintain these patterns farther from an initial natural sequence and whether the improvement follows a predictable scaling law.

DNA language models can extend a short DNA sequence by repeatedly predicting and sampling the next base. Larger models generally predict natural DNA more accurately, but long generated sequences can still differ from real genomes. A sequence may look plausible near its starting point and gradually lose the patterns found in natural DNA. Understanding whether model size delays this deterioration would help determine what scaling achieves beyond better next-base prediction.

Here, realistic DNA means DNA that resembles natural sequences in measurable ways: the frequency of short base patterns, the lengths and spacing of repeats, and the arrangement of gene-coding regions. We will also test whether a classifier can tell generated DNA from natural DNA. These tests measure resemblance to a chosen organism's DNA; they do not establish that a sequence would function in a cell.

2 Prior Work and What Remains Open

Nguyen et al. introduced Evo and showed that DNA prediction improves with increasing model scale across several architectures [1]. Evo could generate long sequences with plausible features, but those sequences still lacked important genomic components. Brixi et al. developed Evo 2 and reported improvements in prediction and generation, while also noting that generated genomes can lack essential elements [2]. These results show that larger models can improve generation, but they do not establish how the distance over which genomic patterns are preserved scales with model size.

Tzanakakis et al. examined the differences between generated and natural DNA, finding problems with repeats, short sequence patterns, regulatory motifs, and organization over long distances [3]. Their accompanying software includes a convolutional neural network (CNN) that distinguishes natural from generated sequences and tests how its performance changes with distance from the initial prompt [4]. We will adapt this evaluation rather than present the classifier or the observation of deterioration as new contributions.

The question we aim to answer is whether different model sizes produce different rates of deterioration, which genomic patterns benefit from scaling, and which remain difficult to preserve. The main result will be a comparison of generation quality across model sizes and distances. Prediction loss will serve as a secondary reference: if quality follows the same trend as loss, that is useful evidence, but reproducing the known loss-scaling relationship is not the project's goal.

3 Plan

Compare models under the same conditions. We will begin with available Evo 2 models, aiming to include the 1B, 7B, and 40B base versions as computing resources allow. We will use the same natural DNA prompts, sampling settings, and amount of preceding context for every model. We will record differences in training data and training duration, since these could also explain differences between released models. We will begin with one annotated bacterial genome and later test whether the findings hold in other organisms.

Measure quality at increasing distances. From each generated continuation, we will extract equal-length windows at increasing distances from the prompt. Each window will be compared with natural DNA from corresponding positions after the same prompt. Using equal-length windows prevents a classifier from gaining an advantage simply by seeing more bases. We will reserve separate genomic regions for training and testing, keep windows from the same region together, and group closely related regions when possible.

We will adapt the published 1D CNN, whose filters scan along DNA to recognize sequence patterns. We will measure how reliably it separates natural and generated windows. Alongside this score, we will compare the lengths and spacing of repeated sequences and the fraction and spacing of predicted gene-coding regions. These measurements will help explain what the classifier detects. Simple comparisons of GC content and short base-pattern frequencies will test whether the difference is mostly a change in composition. A control that compares two matched sets of natural DNA should give near-chance classification.

Test how generation quality scales. For each measure, we will plot quality against distance from the prompt for each model size. We will ask whether larger models deteriorate more slowly and whether the benefit differs across genomic patterns. Where the curves cross a chosen threshold, we will estimate how far each model can generate before crossing it and check whether the result depends strongly on that threshold. We will compare a power-law fit with simpler alternatives, such as improvement that levels off, and report uncertainty across genomic regions and repeated generations. A few model sizes can suggest a trend but cannot establish a general scaling law; additional released training snapshots will be used if available. We will also report prediction loss on the same natural sequences to check whether the generation results reveal a different pattern from prediction alone.

4 First Experiment, Expected Results, and Alternatives

Our first experiment will use 100 separate genomic regions, 1,000-base prompts, and 16,000-base continuations from two model sizes. We will evaluate 2,000-base windows at several distances. This pilot will establish the computing cost and test whether the classifier and repeat measurements reveal a clear change with distance before we expand the study.

We expect generated DNA to become easier to distinguish from natural DNA farther from the prompt. We hypothesize that larger models will delay this change, but that some properties, such as repeat organization, may improve less than others. If the pilot supports this pattern, we will add model sizes, longer continuations, and additional organisms.

If the classifier only detects GC content or short base-pattern frequencies, we will focus on the separate structural measurements and repeat the comparison with composition-matched natural DNA. If the natural-versus-natural control is distinguishable, we will fix the data matching before interpreting generated sequences. If quality deteriorates immediately, we will examine shorter distances; if it stays constant, we will test longer continuations. If size produces little improvement, or too few models are available for a reliable scaling fit, the result will be a measured account of which genomic patterns scaling fails to preserve.

References

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[4] Georgakopoulos-Soares Lab. Synthetic Genomes Benchmark. Software accompanying Tzanakakis et al. Accessed October 17, 2026. `https://github.com/Georgakopoulos-Soares-lab/Synthetic_genomes_benchmark`.